

Abhijit Rajkumar

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WORK EXPERIENCE

AI / ML Intern

09/2025 – 04/2026

Euron (Engagesphere Technology Pvt Ltd) 

Remote

Built MediRAG (RAGnosis), a full-stack AI-powered medical document analysis system using Retrieval-Augmented Generation (RAG).

- Designed end-to-end pipeline: document ingestion → embeddings (LlamaIndex, HuggingFace) → vector search (Pinecone/FAISS) → LLM response generation.
- Developed scalable architecture using FastAPI backend, Next.js frontend, and Docker Compose for deployment.
- Implemented semantic search and source-grounded responses for accurate medical Q&A using LLMs.

Apprentice

01/2022 – 11/2023

Pwskills

Bengaluru, Karnataka

- Led development of machine learning solutions for EDA (Exploratory Data Analysis) products, focusing on innovative and efficient algorithms.
- Collaborated with cross-functional teams including data scientists and software developers to integrate ML capabilities into EDA tools.
- Enhanced problem-solving skills by actively engaging in troubleshooting and resolving complex engineering challenges.

AI / ML Intern

06/2021 – 08/2021

Elite Techno Group

Jaipur, Rajasthan

- Developed a Breast Cancer Prediction model achieving a classification accuracy of 99.45%.
- Engineered a modular ML pipeline with distinct components for data ingestion, transformation, model training, and prediction.
- Gained hands-on experience in end-to-end machine learning workflow development and optimization.

EDUCATION

Bachelor of Science in Mathematics

06/2013 – 06/2017

Gurucharan College

Silchar, Assam

- Core Subjects: Linear Algebra, Real Analysis, Number Theory, Complex Analysis, Mathematical Modeling.
- Applied mathematical concepts to real-world problems in traffic control and rainfall prediction.

TECHNICAL SKILLS

Languages

Python, TypeScript, JavaScript, C, C++, R

AI & LLMs

LangChain, OpenAI GPT, Google Gemini, Inngest Agent Kit, RAG, Pinecone, Qdrant, HuggingFace

Databases

PostgreSQL, MongoDB, MySQL, SQLite, Pinecone, Qdrant, Prisma

Cloud & DevOps

AWS (EC2, S3, Lambda, RDS), GCP (BigQuery), Docker, Kubernetes, GitHub Actions, Nginx

Full-Stack

Next.js, React, FastAPI, Node.js, Express.js, Tailwind CSS, REST APIs, Streamlit

ML

Supervised & Unsupervised Learning, Neural Networks, CNNs, Random Forest, XGBoost, Time Series

Big Data

Hadoop, PySpark, Apache Kafka, Hive, HDFS, AWS EMR

Tools

E2B Sandboxes, Modal, Inngest, Better Auth, Clerk, Polar, Biopython, BeautifulSoup, Postman

PROJECTS

GitHawk

Next.js, Better Auth, Pinecone, Gemini AI, Inngest, Polar, PostgreSQL

GitHub | Live Demo

- Built a full-stack AI-powered GitHub Pull Request reviewer — an open-source alternative to CodeRabbit — using Next.js, Better Auth (GitHub OAuth), and PostgreSQL via Prisma.
- Implemented a RAG pipeline using Pinecone vector embeddings to index entire repository codebases and retrieve semantically relevant context for every PR review.
- Engineered an Inngest background job that fetches the PR diff, runs vector search, generates a structured review with Gemini 2.5 Flash (walkthrough, Mermaid diagram, bug analysis, suggestions), and posts the result as an inline GitHub PR comment — with concurrency control and automatic retries.
- Integrated Polar for subscription management and built a separate index-repo Inngest function to embed all repo files into Pinecone on first repository connection.

vo-clone

Next.js, Inngest Agent Kit, GPT-4.1, E2B Sandboxes, Clerk, PostgreSQL

- Built a full-stack AI UI generation platform inspired by vo.dev — users describe a feature in plain English and receive a live, interactive Next.js preview running inside a real cloud sandbox.
- Architected an Inngest Agent Kit coding agent (GPT-4.1-mini) with terminal, createOrUpdateFiles, and readFiles tools that autonomously scaffolds, writes, installs dependencies, and debugs code inside E2B cloud sandboxes.
- Implemented sandbox reuse logic that reconnects to an existing E2B sandbox across conversation turns, preserving installed node modules and prior file state to eliminate redundant reinstalls.
- Persisted every generation as a Fragment (file snapshot + live URL) linked to a project message thread, enabling full multi-turn iterative refinement with complete history via Prisma and PostgreSQL.

Patho-Predict

Next.js, Modal, FastAPI, Biopython, UCSC API, ClinVar, Pinecone

- Developed a full-stack genomic variant pathogenicity prediction platform — input a chromosomal position and alternative nucleotide, receive an AI-powered clinical classification with evidence.
- Built a Modal serverless FastAPI endpoint that fetches the reference base from the UCSC Genome Browser API, translates the codon change into an amino acid substitution with Biopython, and scores the variant across three signals: CADD-like transition/transversion penalty, biochemical AA severity, and GC-context conservation.
- Designed a weighted scoring model ($\text{risk} = 0.45 \times \text{CADD} + 0.35 \times \text{ESM-proxy} + 0.20 \times \text{Conservation}$) with thresholded output into Likely Pathogenic, Uncertain Significance, or Likely Benign.
- Integrated ClinVar variant overlays, gene sequence visualization, and 7 Next.js API routes consuming UCSC and NCBI APIs — supporting any UCSC genome assembly (hg38, hg19, mm39, etc.).

CERTIFICATIONS

Big Data Engineering Intern

Euron (Engagesphere Technology Pvt Ltd) Sep 2025 – Apr 2026 Remote

- Hadoop, Spark (RDDs, DataFrames, Catalyst), Kafka, Spark Streaming. Orchestration with Airflow and Azure Data Factory.
- Cloud: Azure (Data Lake, Databricks), AWS (S3, EMR, Glue, Athena). Projects: Fraud Detection, Sentiment Analysis.

Data Science Masters Certification

PW Skills Nov 2023 Remote

- End-to-end ML workflows; Deep Learning with TensorFlow, Keras, PyTorch; PCA, t-SNE, LDA for feature engineering.
- Domain projects in healthcare, finance, and retail using Random Forest, XGBoost, SVM, and Logistic Regression.

Full Stack Developer Cohort (0 to 100)

Instructor: Harkirat Singh July 2024 Remote

- MERN stack, System Design, DevOps (Docker, Kubernetes, CI/CD, Nginx, Prometheus, Grafana, AWS).
- Advanced: Kafka, WebSockets, gRPC, Rate Limiting. Projects: Paytm Clone, Zerodha Clone, Zapier Clone.

SCHOLASTIC ACHIEVEMENTS

- Completed an MSc project on "Traffic Control Optimization using Linear Algebra," demonstrating expertise in mathematical modeling. Actively working towards publication.

- Completed an MSc project on rainfall prediction using Time Series analysis, applying advanced statistical techniques to real-world challenges. Pursuing academic publication of findings.